NATIONAL UNIVERSITY
of Computer & Emerging Sciences

MACHINE LEARNING FOR DATA SCIENCE

QUIZ 1

Name:______________________Reg No:_________________________

CLO: 1 **Apply supervised, unsupervised, learning techniques to solve classification and clustering problems of moderate complexity.**

Q No. 1: A telecom company wants to segment its customers based on usage behavior to design targeted service plans. The company has collected data for 15 customers, each described using the following three numerical features:

1. Monthly Call Duration (minutes)
2. Monthly Data Usage (GB)
3. Number of Customer Service Complaints (per month)

Customer	Call Duration (min)	Data Usage (GB)	Complaints
C1	120	1.5	4
C2	300	8.0	1
C3	250	6.5	2
C4	80	0.8	5
C5	200	5.0	3
C6	180	4.2	2
C7	90	1.0	4
C8	320	9.0	1
C9	260	7.2	1
C10	70	0.5	6
C11	210	5.5	3
C12	140	2.0	4
C13	290	8.5	1
C14	160	3.0	3
C15	100	1.2	5

Assume the initial centroids are chosen as the normalized feature vectors of **C2, C4, and C10.**

A. Apply Standard Normalization (Z-score normalization) to the dataset.

B. Using the normalized data, apply K-Means clustering with k=3.

C. Compute the new centroids after the first iteration of K-Means.

ADVANCE MACHINE LEARNING

QUIZ 1

Name:________________________Reg No:____________________________

CLO: 1 **Apply supervised, unsupervised, learning techniques to solve classification and clustering problems of moderate complexity.**

Q No. 1: A bank wants to classify whether a customer is “High Risk” or “Low Risk” for loan approval based on historical behavioral data. The bank has collected data from 15 previous customers, each described using the following three numerical features:

1. Annual Income (in thousand USD)
2. Credit Utilization Ratio (%)
3. Number of Late Payments (per year)

Customer	Income (k$)	Credit Utilization (%)	Late Payments	Risk Class
C1	30	75	6	High
C2	85	30	1	Low
C3	60	45	2	Low
C4	28	80	7	High
C5	55	50	3	Low
C6	40	65	4	High
C7	90	25	1	Low
C8	70	40	2	Low
C9	35	70	5	High
C10	25	85	8	High
C11	65	48	2	Low
C12	45	60	4	High
C13	95	20	0	Low
C14	50	55	3	Low
C15	32	72	6	High

A new loan applicant (X) has the following profile:

- Income: 58 k$
- Credit Utilization: 52%
- Late Payments: 3

1. **Apply Standard Normalization (Z-score normalization) to all 15 training instances, also Normalize the new applicant (X) using the same mean and standard deviation values.**
2. **Using the normalized data, apply the KNN algorithm with: k=3 & 5.**
3. **Perform majority voting among the nearest neighbors and determine the predicted risk class for the new applicant.**